Doublecortin is a structural marker for Taxol-induced expansion and compaction of the microtubule lattice

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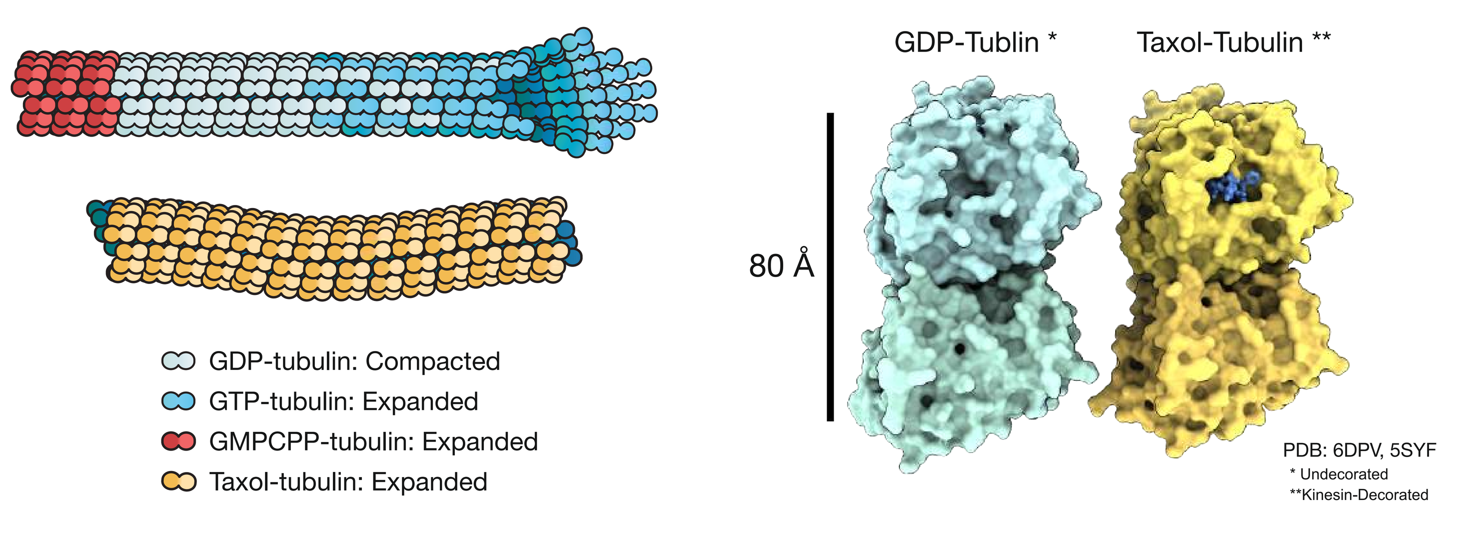
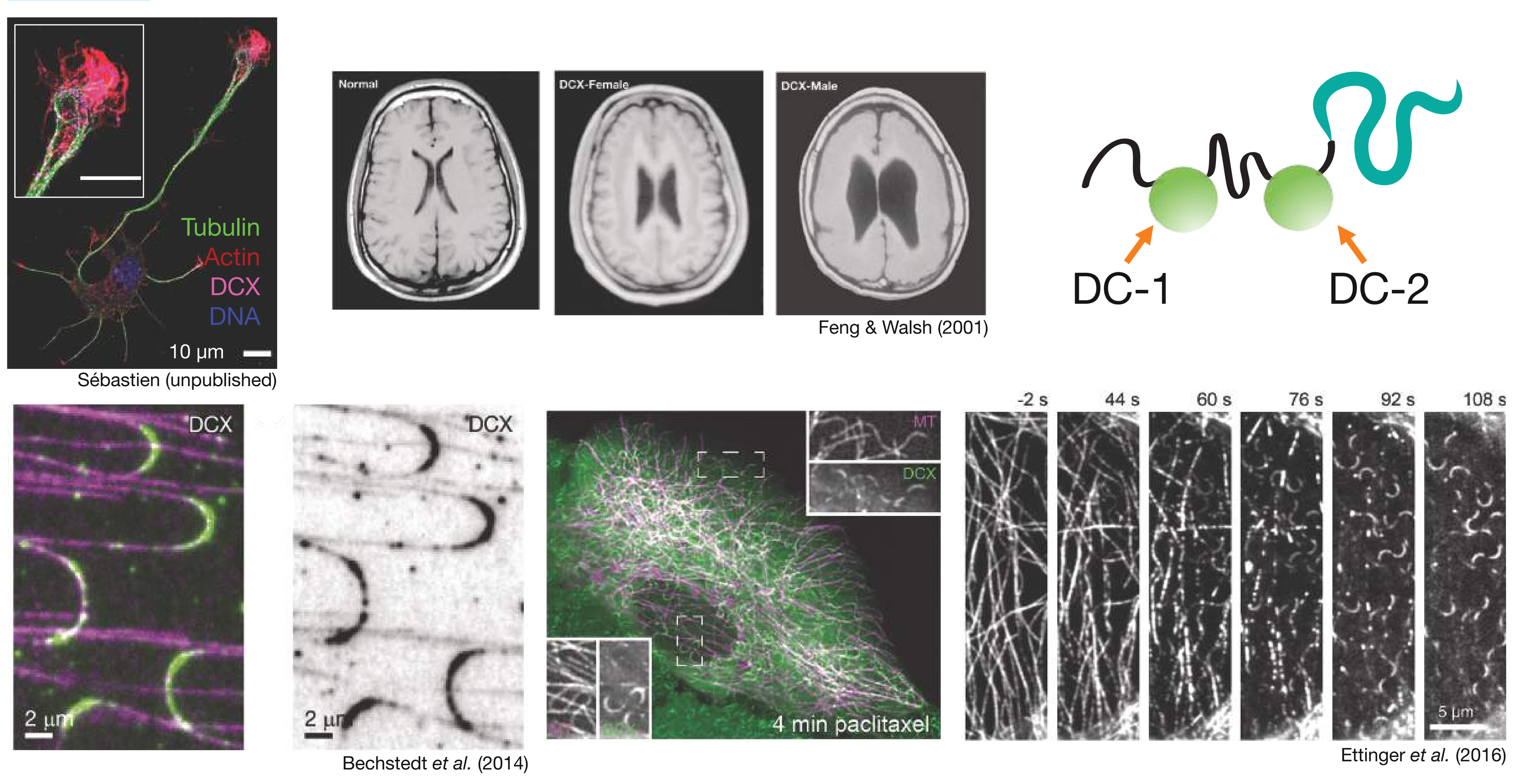
Take-home message

- Taxol-induced microtubule lattice expansion alters the affinity of the MAP DCX.
- Clinically relevant Taxol concentrations cause DCX accumulation into puncta, which may indicate local alteration of lattice conformation.
- High Taxol concentrations cause DCX relocalization to curved microtubules, even when the DCX curvature-recognition domain is mutated.

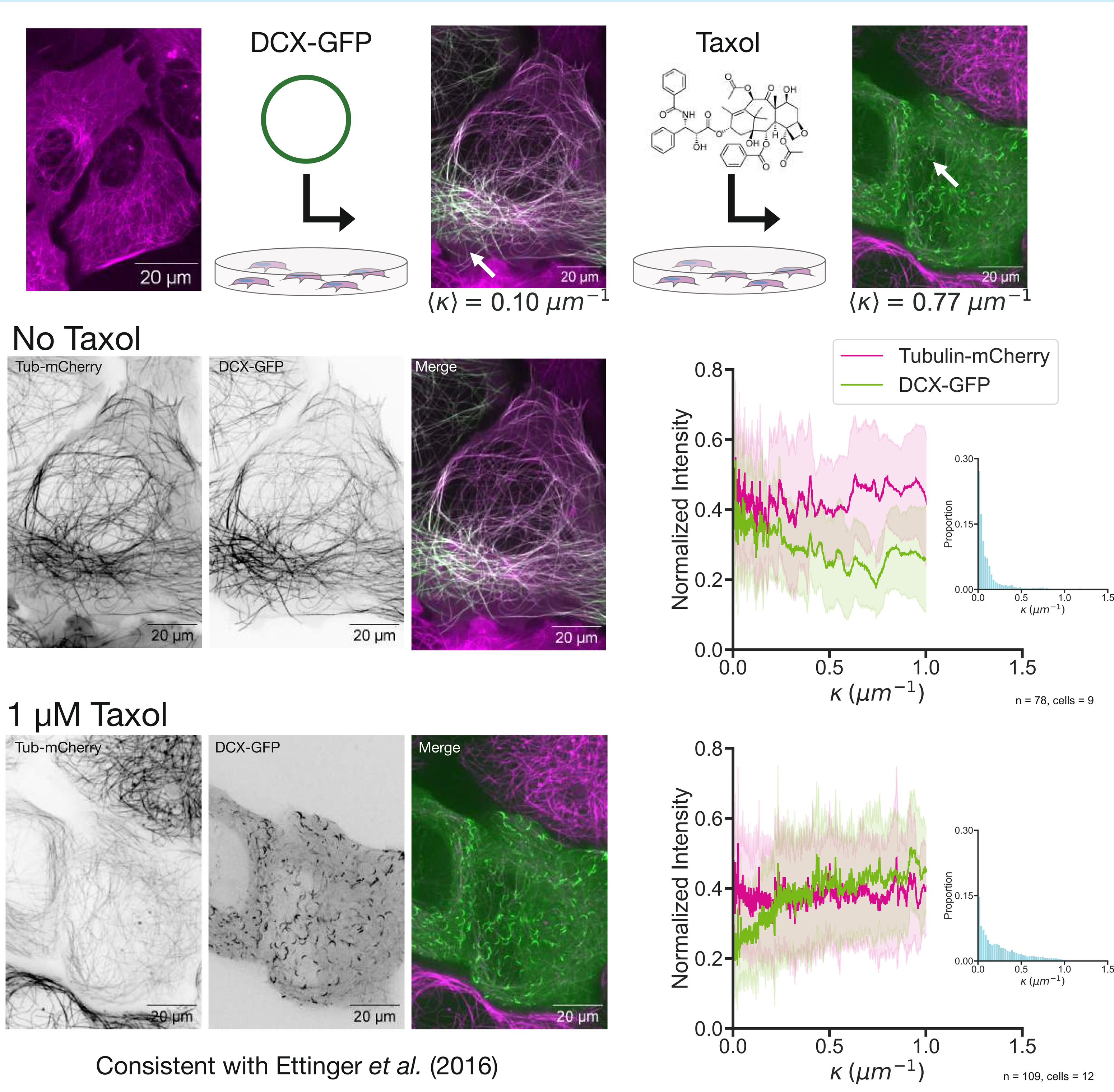
Microtubules expand & compact



DCX localizes to curves

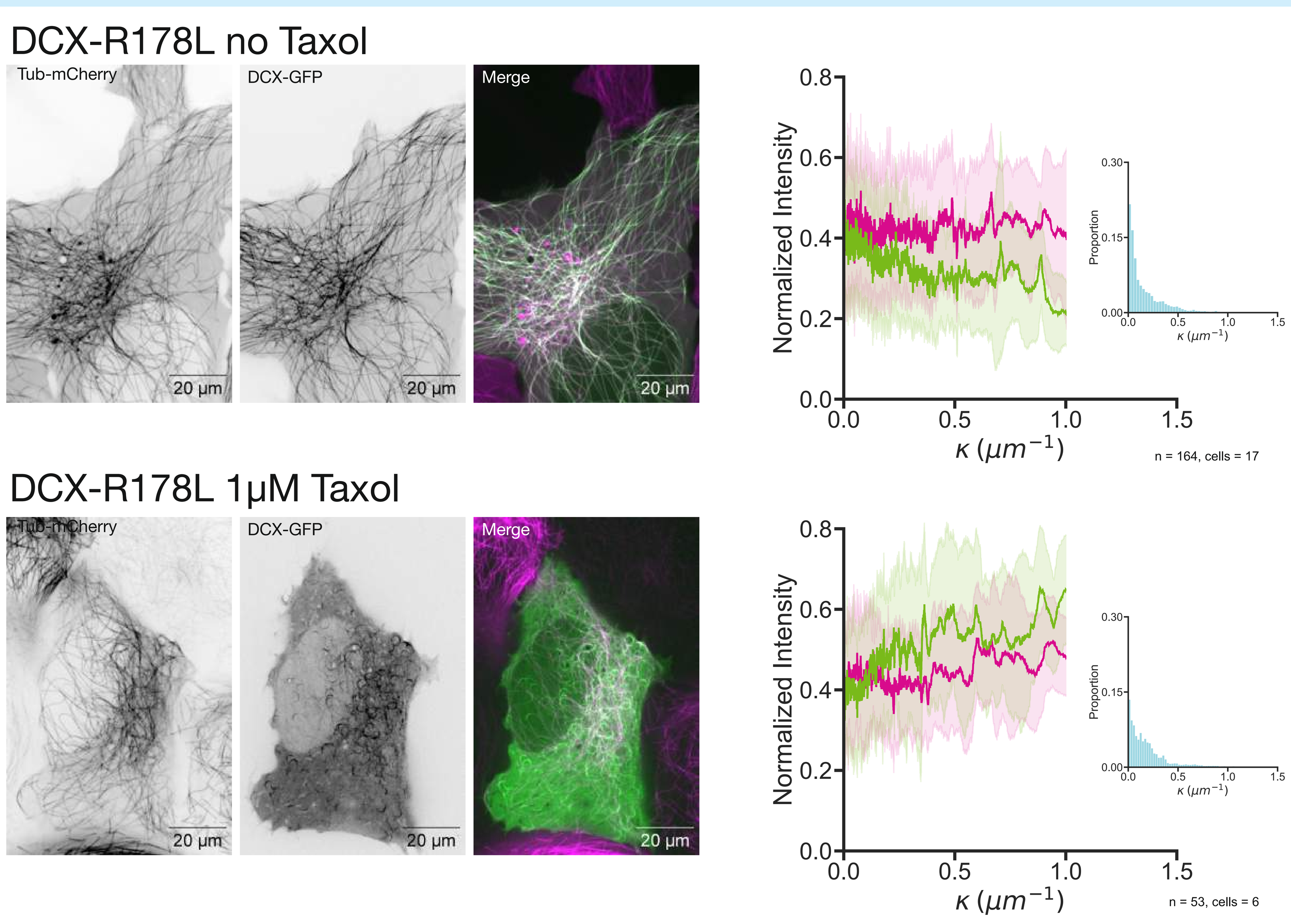


DCX localization changes at high Taxol concentrations

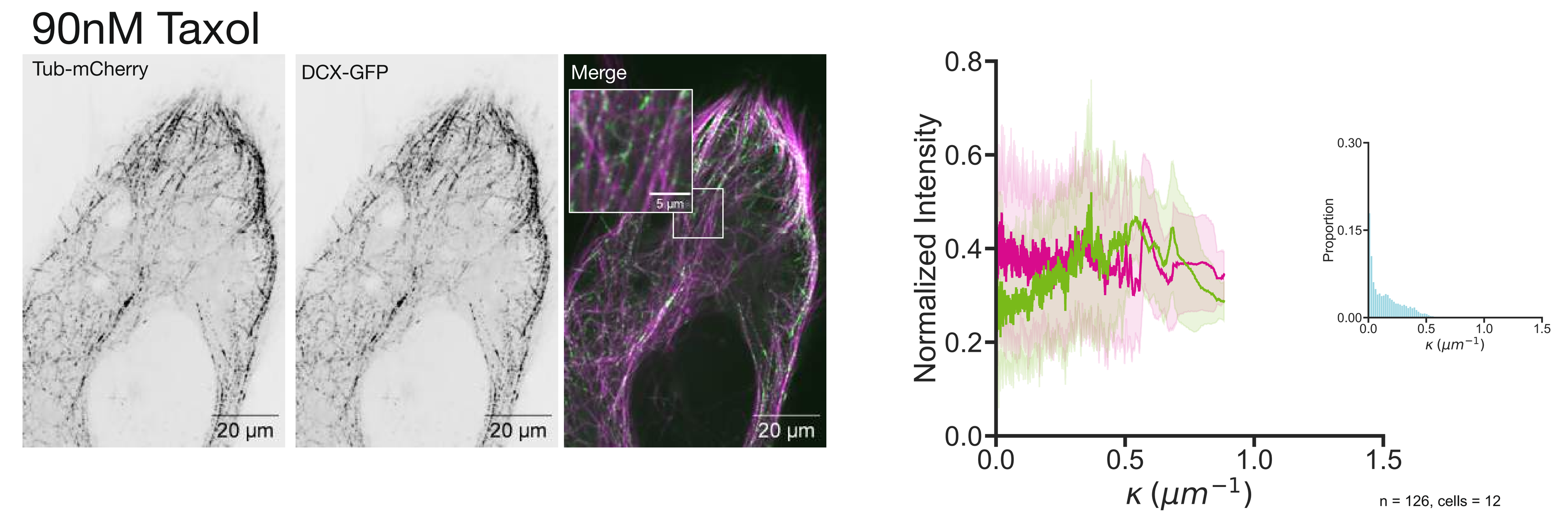


Consistent with Ettinger *et al.* (2016)

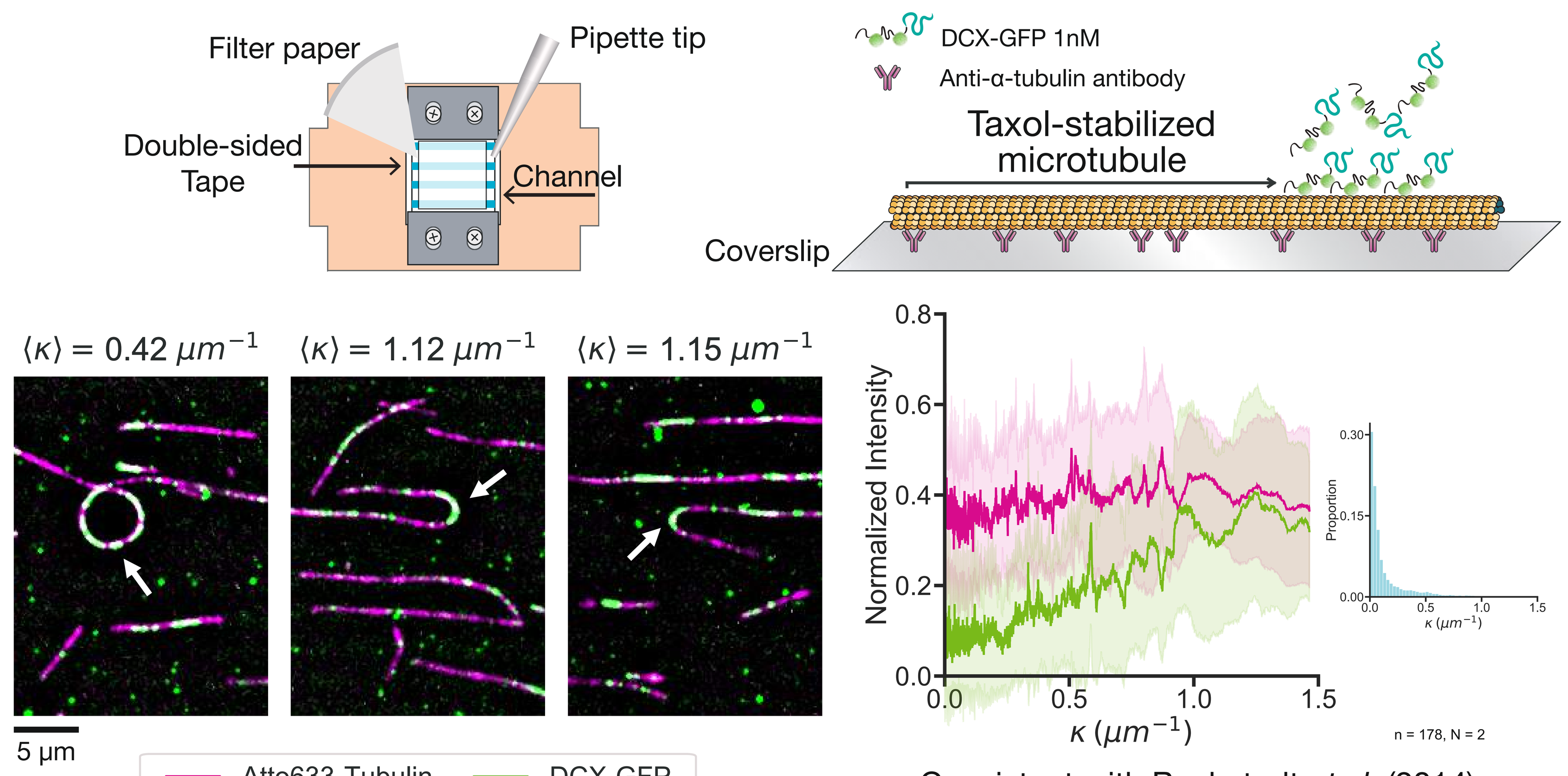
DCX curvature-recognition mutant localizes to curves with Taxol



DCX forms puncta at clinically-relevant Taxol concentrations

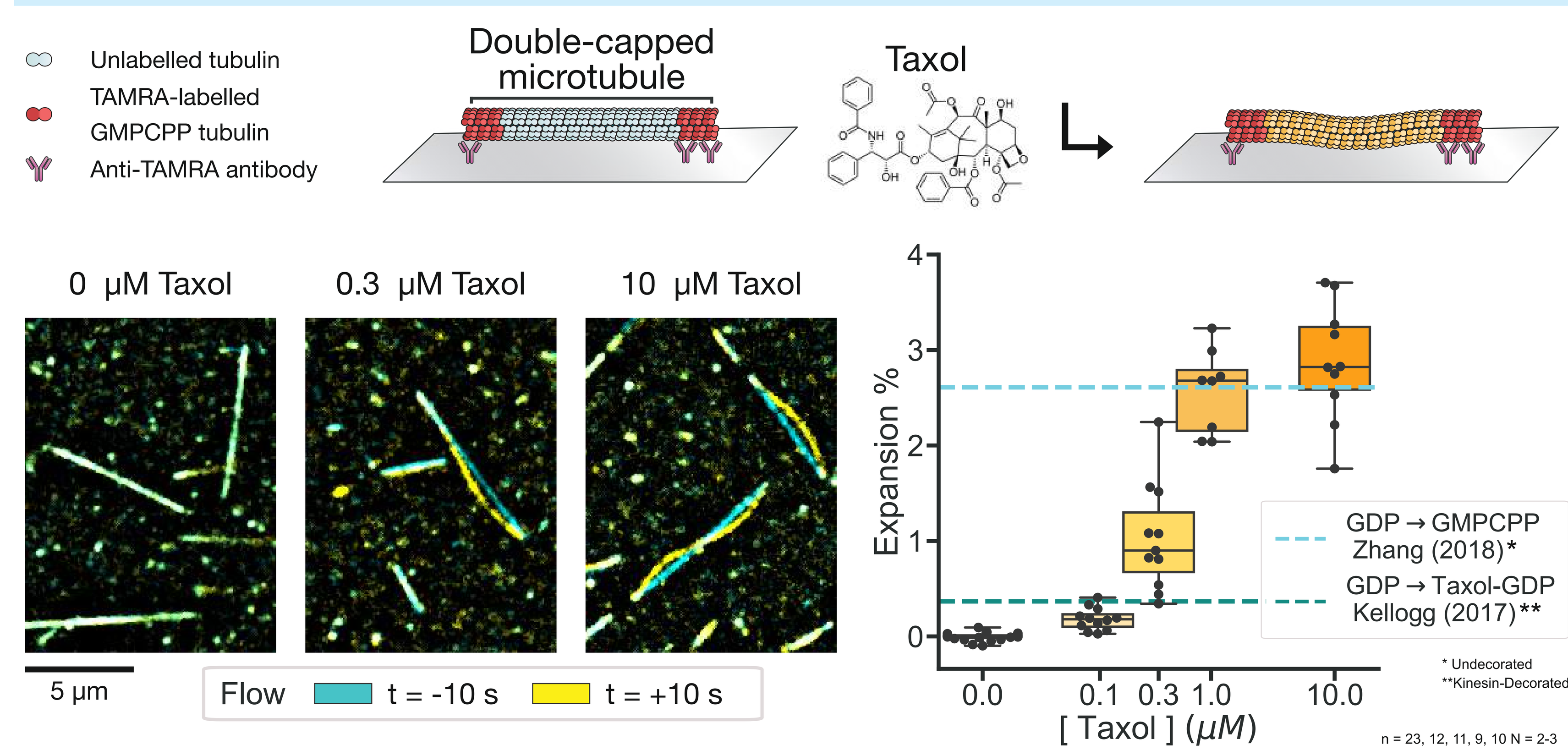


DCX prefers the curved Taxol-expanded lattice *in vitro*

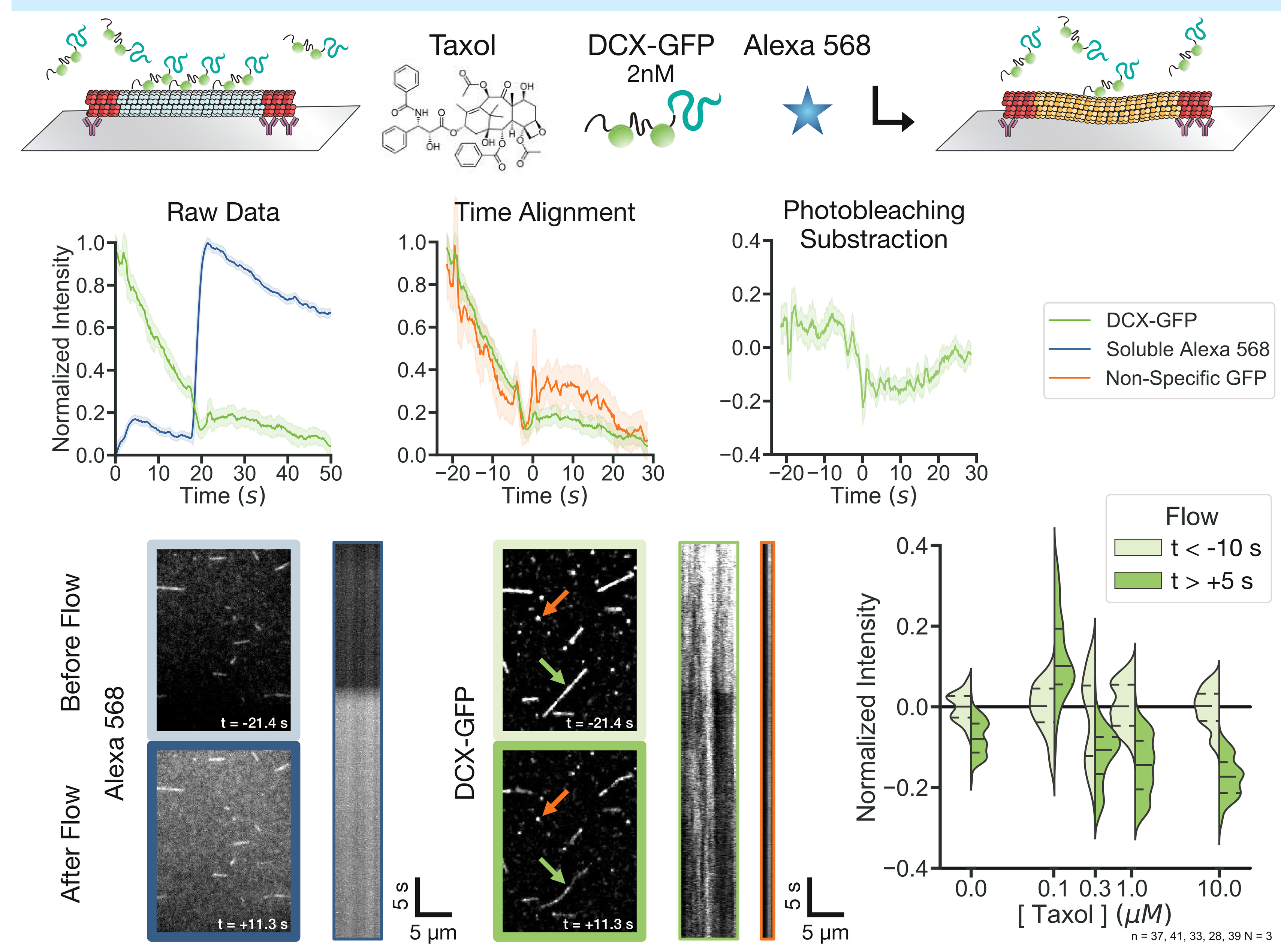


Consistent with Bechstedt *et al.* (2014)

Taxol-induced microtubule expansion is cooperative



Microtubule lattice expansion reduces DCX affinity



Future examinations

- Changes in the localization of other MAPs as a response to other taxane-site stabilising agents *in vivo*
- Effect of Taxol on post-translational modifications and the potential link to DCX localization
- Response of DCX cooperativity to Taxol treatment *in vitro*

References

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